Course Code: PHY2005	Game Physics Development(Physics Engine)	Course Type P Credit 2

Objective

Physics Engine is a lab course for the students learning game programming. The course will train the students to visualize game actions by imbibing physics concepts during game designing process. Also to understand common actions in gaming in terms of the mathematical and physical properties

Expected outcome

- Using physics engine for realistic games.
- Create physics bodies
- Simulated the bodies with physical properties
- Create interactions / collision with the bodies.
- Develop fascinate games attracted with realistic physics concepts.

Expt. No.	Title of the Experiments	Total Lab Hrs.	SLO
1	Vectors : scalar multiplication, addition and subtraction	45	b,c,g, k, l
2	Linear motion of objects (multi-dimension) – variable speed.		
3	Motion of object (e.g. cube, ball etc.) and interaction with static object (e.g. Wall)		
4	Two object motions – Same and opposite direction.		
5	Falling down of objects - gravity effect		
6	Bouncing of objects.		
7	Projectile motion of objects.		
8	Collision of one mobile and one static object.		
9	Collision of any two objects in motion.		
10	Collision of multiple objects		
11	Penetration of objects / bodies		
12	Sound mixing effects		

Recommendation by the Board of Studies on	*11th July 2018*
Approval by Academic council on	*18th July 2018*
Compiled by	*Dr. T S Pradeep Kumar*